



# Analisis Data Mobil: Korelasi antara Engine Size dan Highway MPG terhadap Price

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# Dataset Mobil

- Mencakup simboling, fuel-type body-style, engine-size, horse power, dan lainnya
- Alfa-Romeo, Audi, BMW, Chevrolet, Dodge, Honda
- Engine-size, Highway-mpg, dan price sebagai focus analisis

| I               | J          | K      | L     | M      | N           | O           |
|-----------------|------------|--------|-------|--------|-------------|-------------|
| engine-location | wheel-base | length | width | height | curb-weight | engine-type |
| front           | 88.6       | 168.8  | 64.1  | 48.8   | 2548        | dohc        |
| front           | 88.6       | 168.8  | 64.1  | 48.8   | 2548        | dohc        |
| front           | 94.5       | 171.2  | 65.5  | 52.4   | 2823        | ohc         |
| front           | 99.8       | 176.6  | 66.2  | 54.3   | 2337        | ohc         |
| front           | 99.4       | 176.6  | 66.4  | 54.3   | 2824        | ohc         |
| front           | 99.8       | 177.3  | 66.3  | 53.1   | 2507        | ohc         |
| front           | 105.8      | 192.7  | 71.4  | 55.7   | 2844        | ohc         |
| front           | 105.8      | 192.7  | 71.4  | 55.7   | 2954        | ohc         |
| front           | 105.8      | 192.7  | 71.4  | 55.9   | 3086        | ohc         |
| front           | 99.5       | 178.2  | 67.9  | 52     | 3053        | ohc         |
| front           | 101.2      | 176.8  | 64.8  | 54.3   | 2395        | ohc         |
| front           | 101.2      | 176.8  | 64.8  | 54.3   | 2395        | ohc         |
| front           | 101.2      | 176.8  | 64.8  | 54.3   | 2710        | ohc         |
| front           | 101.2      | 176.8  | 64.8  | 54.3   | 2765        | ohc         |
| front           | 103.5      | 189    | 66.9  | 55.7   | 3055        | ohc         |
| front           | 103.5      | 189    | 66.9  | 55.7   | 3230        | ohc         |
| front           | 103.5      | 193.8  | 67.9  | 53.7   | 3380        | ohc         |
| front           | 110        | 197    | 70.9  | 56.3   | 3505        | ohc         |
| front           | 88.4       | 141.1  | 60.3  | 53.2   | 1488        | I           |
| front           | 94.5       | 155.9  | 63.6  | 52     | 1874        | ohc         |
| front           | 94.5       | 158.8  | 63.6  | 52     | 1909        | ohc         |
| front           | 93.7       | 157.3  | 63.8  | 50.8   | 1876        | ohc         |
| front           | 93.7       | 157.3  | 63.8  | 50.8   | 1876        | ohc         |
| front           | 93.7       | 157.3  | 63.8  | 50.8   | 2128        | ohc         |
| front           | 93.7       | 157.3  | 63.8  | 50.6   | 1967        | ohc         |
| front           | 93.7       | 157.3  | 63.8  | 50.6   | 1989        | ohc         |
| front           | 93.7       | 157.3  | 63.8  | 50.6   | 1989        | ohc         |
| front           | 93.7       | 157.3  | 63.8  | 50.6   | 2191        | ohc         |
| front           | 103.3      | 174.6  | 64.6  | 59.8   | 2535        | ohc         |
| front           | 95.9       | 173.2  | 66.3  | 50.2   | 2811        | ohc         |
| front           | 86.6       | 144.6  | 63.9  | 50.8   | 1713        | ohc         |
| front           | 86.6       | 144.6  | 63.9  | 50.8   | 1819        | ohc         |
| front           | 93.7       | 150    | 64    | 52.6   | 1837        | ohc         |
| front           | 93.7       | 150    | 64    | 52.6   | 1940        | ohc         |
| front           | 93.7       | 150    | 64    | 52.6   | 1956        | ohc         |
| front           | 96.5       | 163.4  | 64    | 54.5   | 2010        | ohc         |

# Engine Size & Price

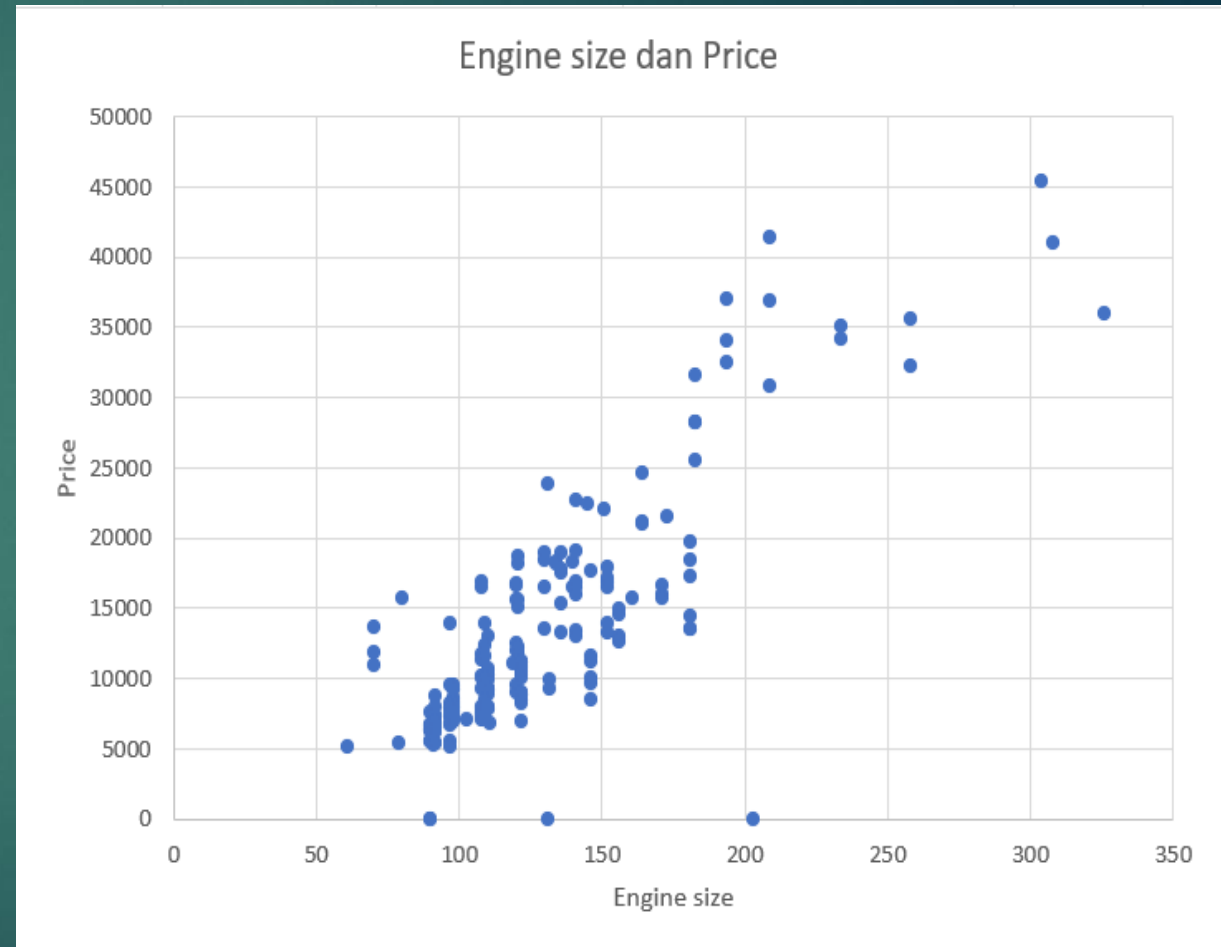
- **Engine-size** — kapasitas mesin dalam cc (rentang 61–209)
- **Price** — harga mobil dalam USD (rentang 5.151–41.315)

Total **27 baris data** digunakan untuk menghitung korelasi dan membuat scatter plot.

|    | A           | B     |
|----|-------------|-------|
| 1  | Engine-size | Price |
| 2  | 130         | 13495 |
| 3  | 130         | 16500 |
| 4  | 152         | 16500 |
| 5  | 109         | 13950 |
| 6  | 136         | 17450 |
| 7  | 136         | 15250 |
| 8  | 136         | 17710 |
| 9  | 136         | 18920 |
| 10 | 131         | 23875 |
| 11 | 131         | ?     |
| 12 | 108         | 16430 |
| 13 | 108         | 16925 |
| 14 | 164         | 20970 |
| 15 | 164         | 21105 |
| 16 | 164         | 24565 |
| 17 | 209         | 30760 |
| 18 | 209         | 41315 |
| 19 | 209         | 36880 |
| 20 | 61          | 5151  |
| 21 | 90          | 6295  |
| 22 | 90          | 6575  |
| 23 | 90          | 5572  |
| 24 | 90          | 6377  |
| 25 | 98          | 7957  |
| 26 | 90          | 6229  |
| 27 | 90          | 6692  |
| 28 | 90          | 7600  |

# Scatter plot

- Scatter plot menunjukkan sumbu X (Engine Size: 0–350) dan sumbu Y (Price: 0–50.000). Titik data membentuk **pola naik** yang jelas.
- Kelompok data padat di engine size 70–200 dengan harga 5.000–25.000
- Beberapa outlier harga tinggi terlihat di engine size ~300–320



## Hasil Korelasi

| C           | D           |
|-------------|-------------|
| Korelasi    | Average     |
| 0.872335167 | 13207.12935 |

Berdasarkan scatter plot dan nilai korelasi, titik-titik data cenderung membentuk pola naik. Hasil perhitungan menggunakan fungsi correl menghasilkan nilai korelasi 0.872335167, yang mempunyai arti korelasi positif kuat atau menandakan kalau engine size memiliki pengaruh yang signifikan terhadap harga mobil dan kedua data saling berhubungan